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PAPER_995: 99-System Gamma Sweep Computation

Abstract

We compute the aggregate F_{U,Bi_i} across all 99 astrophysical systems in the UQFF catalogue at 7 linewidth values $\Gamma \in \{0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0\}$ THz.

1. System Catalogue

The 99 systems span 6 categories: - **Stars** (20): Main sequence, giants, supergiants (0.1 – $100 M_\odot$) - **Galaxies** (20): Spirals through ellipticals (10^9 – $10^{13} M_\odot$) - **Nebulae** (15): Planetary through H II regions - **Compact Objects** (15): Neutron stars (1.4 – $2.5 M_\odot$) and stellar BHs (3 – $100 M_\odot$) - **Clusters** (15): Globular through galaxy superclusters (10^{13} – $10^{16} M_\odot$) - **Cosmological** (14): Large-scale structure (10^{15} – $10^{17} M_\odot$)

2. Aggregate at Reference Linewidth

$$F_U^{(99)}(\Gamma_0 = 0.1 \text{ THz}) = -6.11 \times 10^{13} \text{ m/s}^2$$

The negative sign confirms global inward dominance across all mass scales.

3. Stability

100% of systems return finite, stable F_{U,Bi_i} values across all 7 Γ values — no divergences or instabilities detected.

4. Implementation

File: 99system_wstp_gamma.py, class NinetyNineSystemGammaSweep. CP4 class #579.